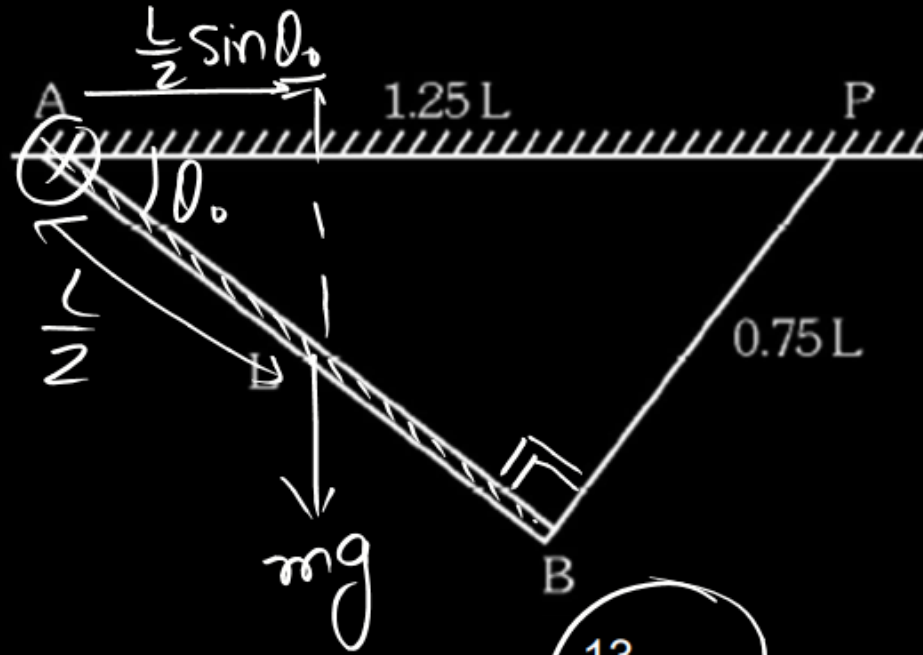


A uniform rod AB of mass m and length L is tied, at its end B, to a thread which is attached to point P on the ceiling. Length of the thread PB is $0.75L$. The other end A of the rod is hinged at a point on the ceiling. Distance $AP = 1.25L$. End B of the rod is pushed gently perpendicular to the plane of the figure and it starts oscillating.



(A) The moment of inertia of the rod about line AP is $\frac{13}{25}mL^2$

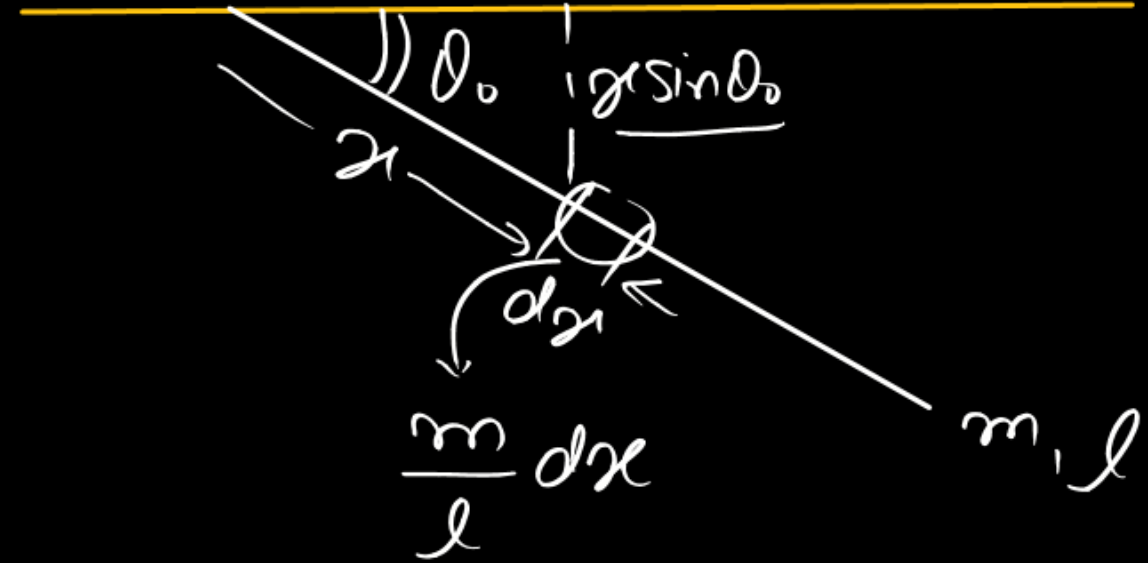
(B) Assuming that the triangle APB makes a small angle θ with the vertical plane. The time period of oscillation is $2\pi\sqrt{\frac{2L}{3g}}$

(C) Assuming that the triangle APB makes a small angle θ with the vertical plane. The time period of oscillation is $2\pi\sqrt{\frac{2L}{5g}}$

(D) The moment of inertia of the rod about line AP is $\frac{3mL^2}{25}$

$$\frac{ml^2}{3} \times \frac{3}{25}$$

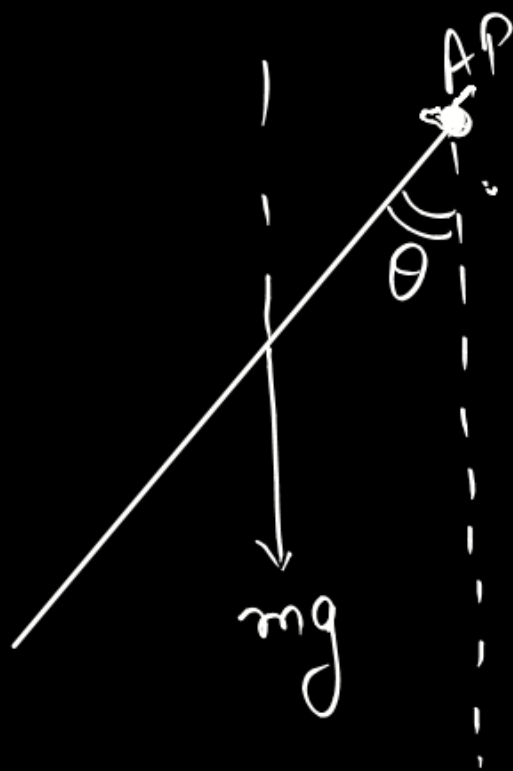
$$\boxed{\frac{3ml^2}{25}}$$



$$dI = \frac{m}{l} dx x^2 \sin^2 \theta_0$$

$$I = \frac{m}{l} \sin^2 \theta_0 \int_0^l x^2 dx$$

$$= \frac{ml^2}{3} \sin^2 \theta_0$$



$$\tau = \left(mg \frac{l}{2} \sin \theta_0 \right) \sin \theta$$

$$\tau = \left(mg \frac{l}{2} \sin \theta_0 \right) \theta$$

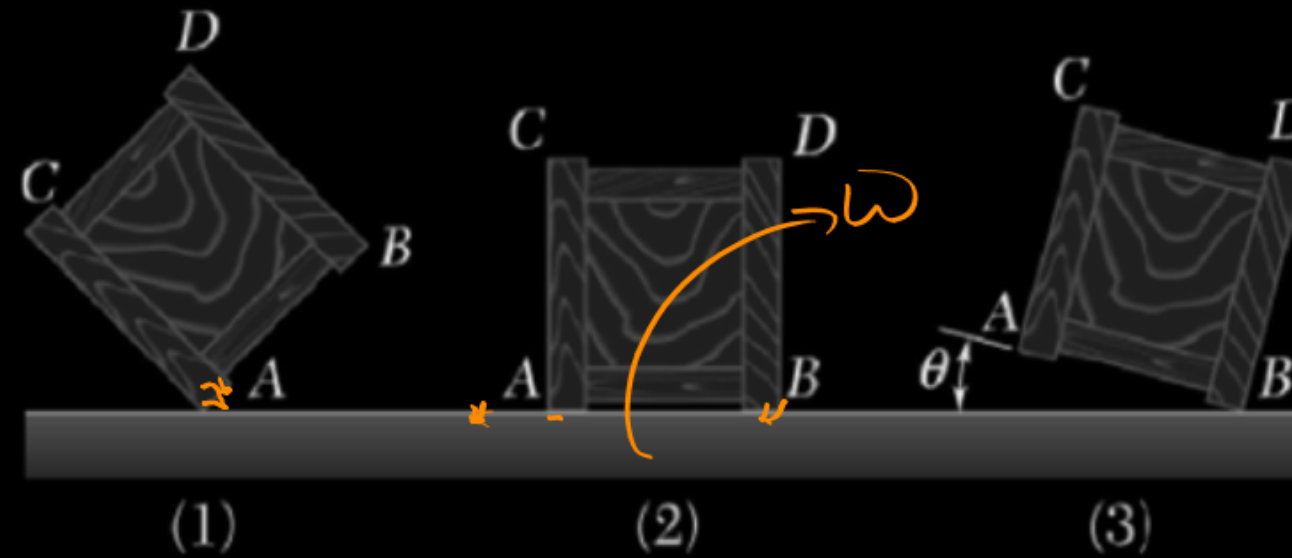
$$= \cancel{m} g \frac{\cancel{l}}{2} \times \frac{3}{5} \theta = \left(\frac{3}{5} \cancel{m} \cancel{l}^2 \right) \alpha$$

$$\alpha = \underbrace{\left(\frac{5g}{2l} \right)}_{\omega^2} \theta$$

$$\omega = \sqrt{\frac{5g}{2l}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{2l}{5g}}$$

A uniformly loaded square crate is released from rest with its corner D directly above A; it rotates about A until its corner B strikes the floor, and then rotates about B. The floor is sufficiently rough to prevent slipping and the impact at B is perfectly inelastic. Denoting by ω_0 the angular velocity of the crate immediately before B strikes the floor.



The angular velocity of the crate immediately after B strikes the floor-

(A) $\frac{\omega_0}{2}$

(B) $\frac{\omega_0}{4}$

(C) $\frac{\omega_0}{8}$

(D) $\frac{\omega_0}{3}$

The fraction of the kinetic energy of the crate lost during the impact-

(A) $\frac{1}{8}$

(B) $\frac{16}{22}$

(C) $\frac{15}{16}$

(D) $\frac{22}{35}$

$$K_i = \frac{1}{2} m v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2 = \frac{1}{2} I_{IAR} \omega_0^2 = \frac{1}{2} \cdot \frac{2}{3} m l^2 \omega_0^2$$

$$= \frac{1}{2} m \left(\frac{\omega_0 l}{\sqrt{2}} \right)^2 + \frac{1}{2} \times \frac{m l^2 \omega_0^2}{6}$$

$$= \frac{1}{4} m \omega_0^2 l^2 + \frac{m \omega_0^2 l^2}{12} = \frac{4}{12} m \omega_0^2 l^2 = \frac{1}{3} m \omega_0^2 l^2 = K$$

$$K_f = \frac{1}{2} \times \frac{2}{3} m l^2 \left(\frac{\omega_0^2}{16} \right) = \frac{m l^2 \omega_0^2}{48} = \frac{K}{16}$$

$$\frac{K_i - K_f}{K_i}$$

$$\frac{K - K/16}{K} = \frac{15}{16}$$

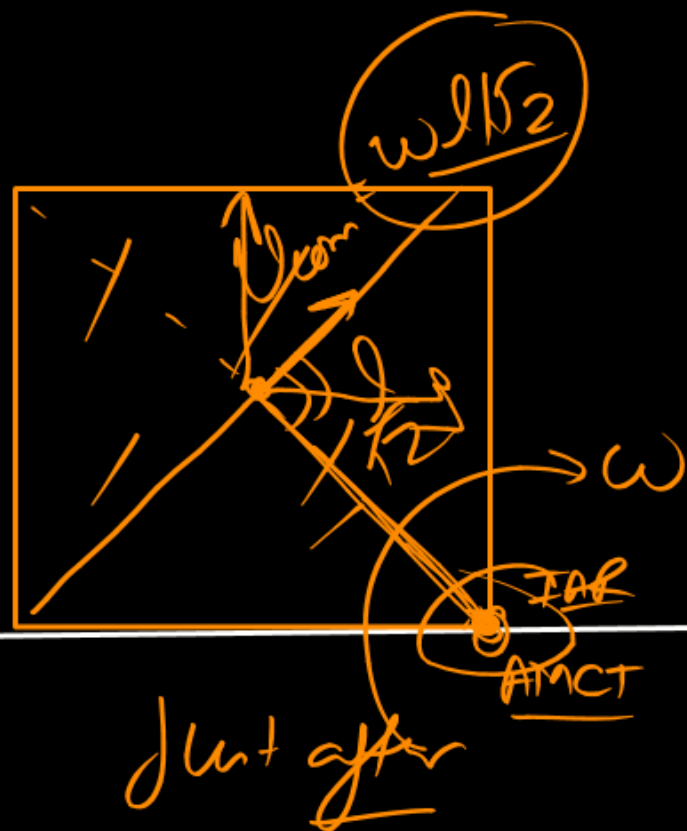
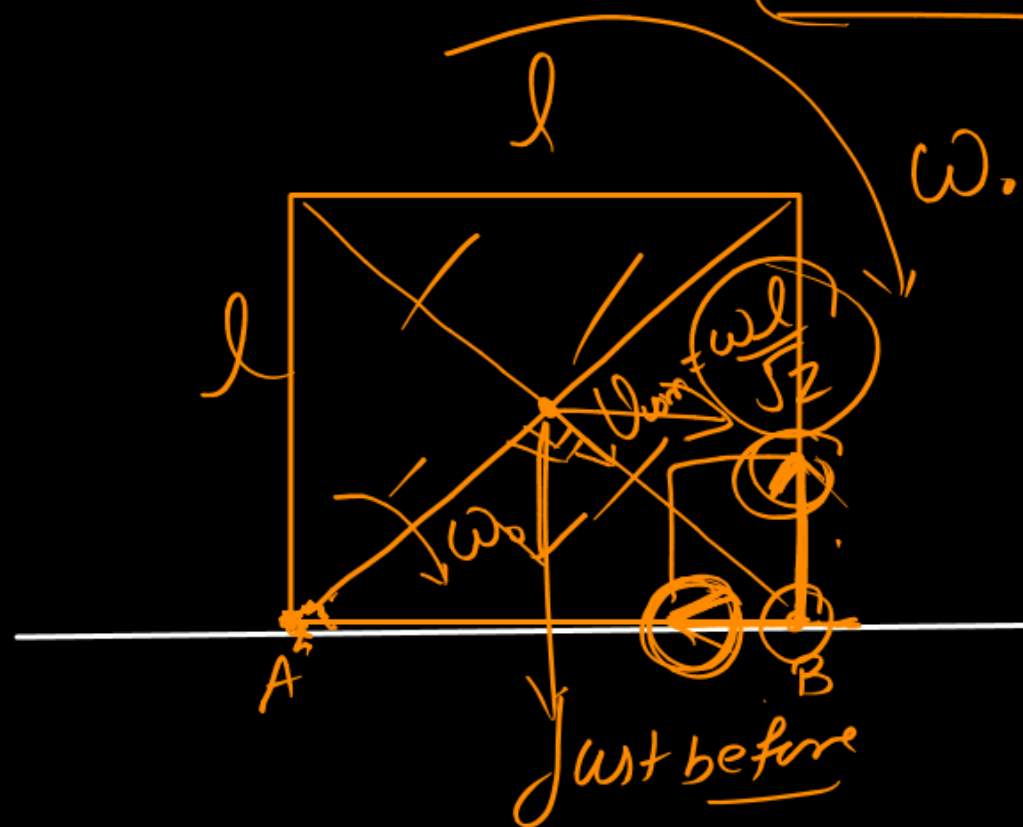
$$\frac{ml^2}{62} \omega_0 = \frac{2}{3} ml^2 \omega$$

$$\omega = \frac{\omega_0}{4}$$

$$m \vec{r} \times \vec{v}_{\text{cm}}$$

$$\vec{L}_i = \frac{ml^2}{6} \omega_0 \curvearrowright + 0$$

$$L_i = \frac{ml^2}{6} \omega_0 \curvearrowright$$

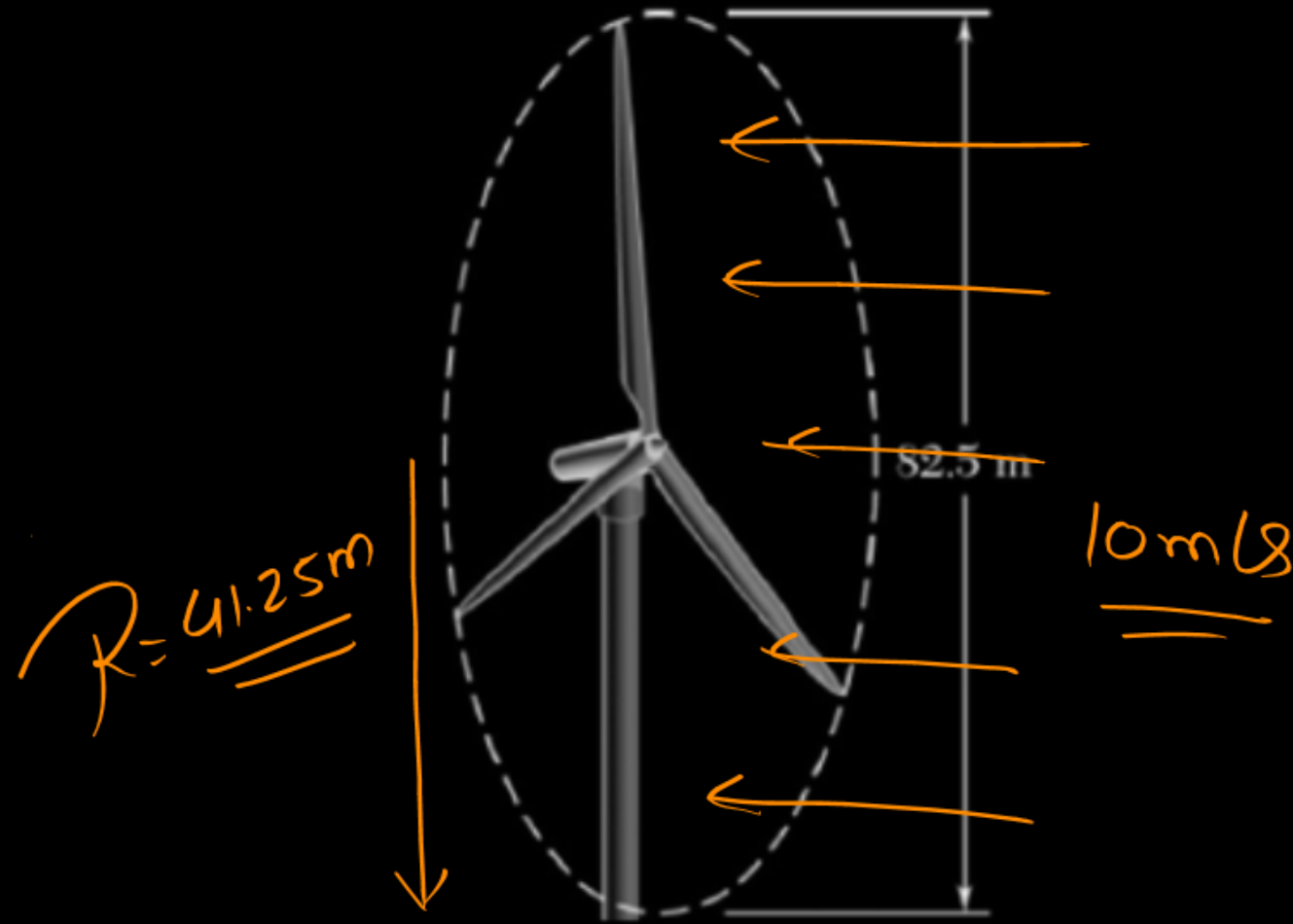


$$\vec{L}_f = I_{\text{PAR}} \omega = \frac{2}{3} ml^2 \omega \curvearrowright$$

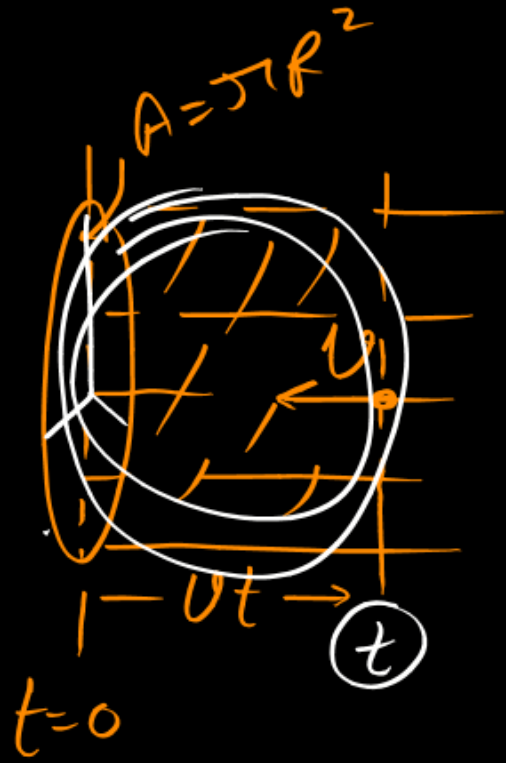
$$\textcircled{I} \quad \vec{L}_f = \frac{ml^2}{6} \omega \curvearrowright + \frac{ml}{\sqrt{2}} \frac{\omega l}{\sqrt{2}} \curvearrowright$$

$$= \frac{ml^2}{6} \omega + \frac{ml^2}{2} \omega = \frac{1+3}{6} ml^2 \omega = \frac{2}{3} ml^2 \omega$$

The wind turbine-generator shown has an output-power rating of 1.5 MW for a wind speed of 36 km/h. For the given wind speed, the kinetic energy of the air particles entering the 82.5-m-diameter circle per second is $(n+0.23)$ MW. Find the value of n (Assume $\rho = 1.21 \text{ kg/m}^3$ for air)



$$\begin{aligned} &1.5 \times 10^6 \text{ Watt} \\ &\underline{\underline{1.5 \times 10^6 \frac{\text{Joule}}{\text{sec}}}} \end{aligned}$$



$$\text{Efficiency} = \frac{1.5}{3.23}$$

$$\frac{K}{t} = \frac{\frac{1}{2} (\rho A U^3) U^2}{K}$$

$$= \frac{1}{2} \rho A U^3$$

$$= \frac{1}{2} \times 1.21 \times \pi \times (41.25)^2 \times (10)^3 \quad \frac{\text{Joule}}{\text{sec}}$$

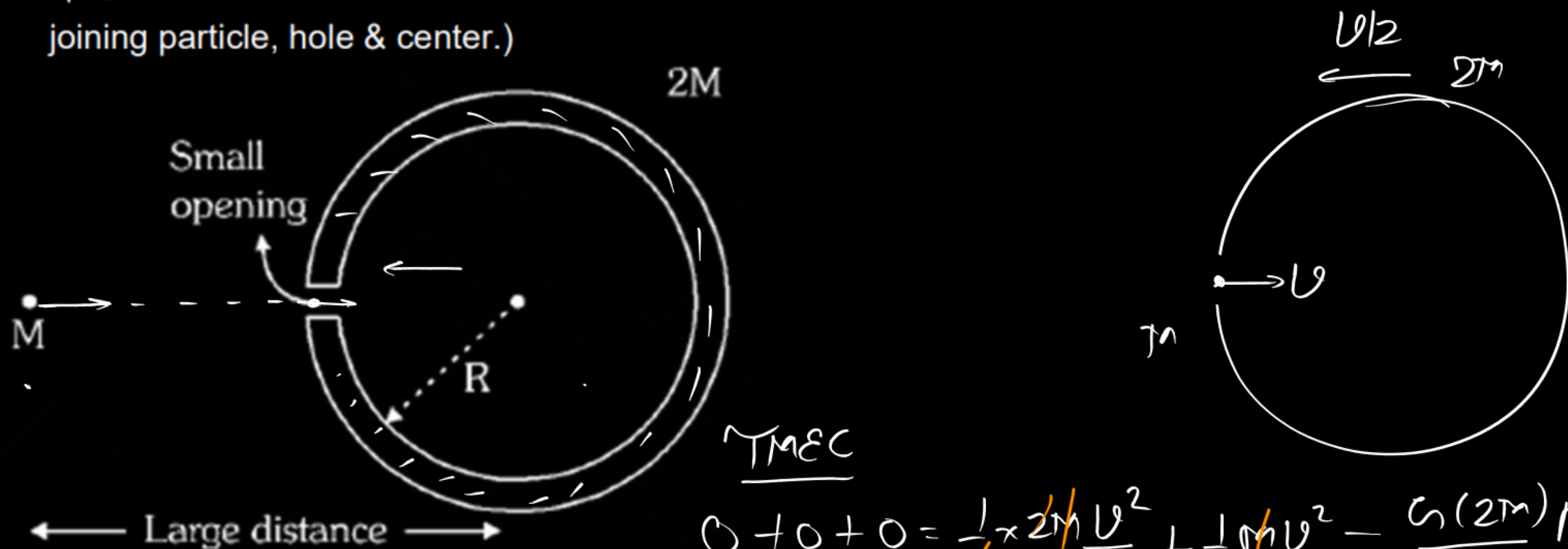
$$= 3.23 \times 10^6 \text{ Watt}$$

$$= 3.23 \text{ MWatt}$$

$$(n+0.23)$$

$$n=3$$

The shown set up is released at rest. The speed of mass M when it just comes out of the shell is $\sqrt{\frac{P GM}{3 R}}$. Find P (Assume all collisions to be elastic and also assume particle will move on the line joining particle, hole & center.)



$$0 + 0 + 0 = \frac{1}{2} \times 2M \frac{U^2}{4} + \frac{1}{2} M U^2 - \frac{G(2M)M}{R}$$

$$U = \sqrt{\frac{8GM}{3R}}$$

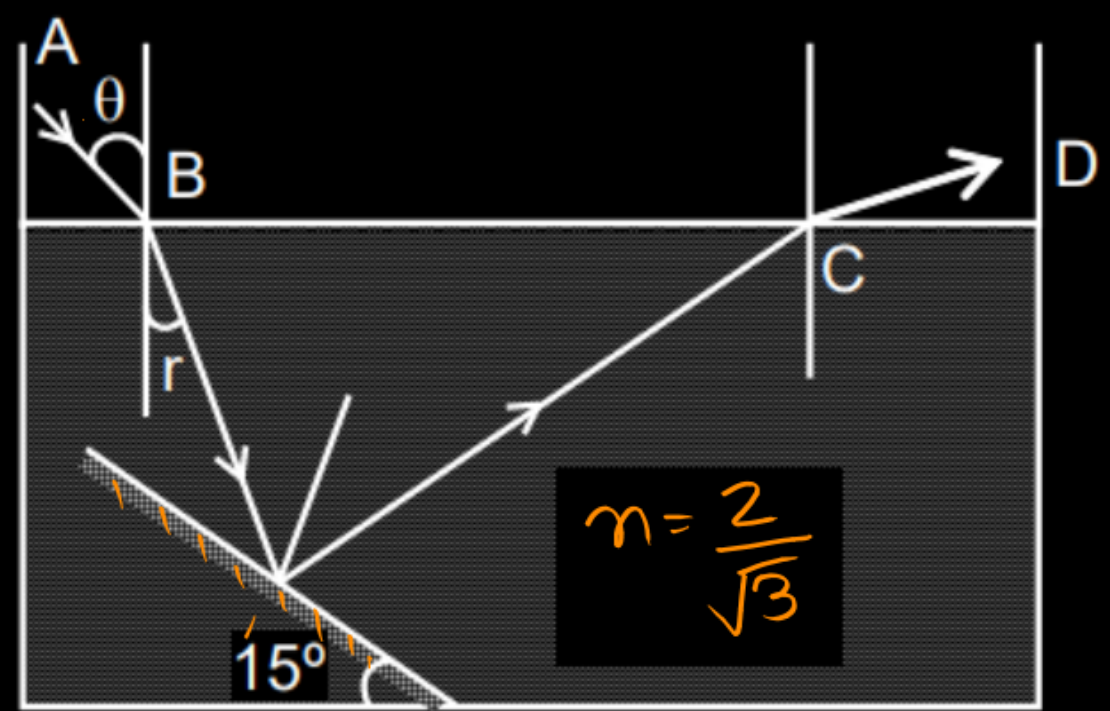
$$\frac{3}{4} U^2 = \frac{2GM}{R}$$

$$P = 8$$

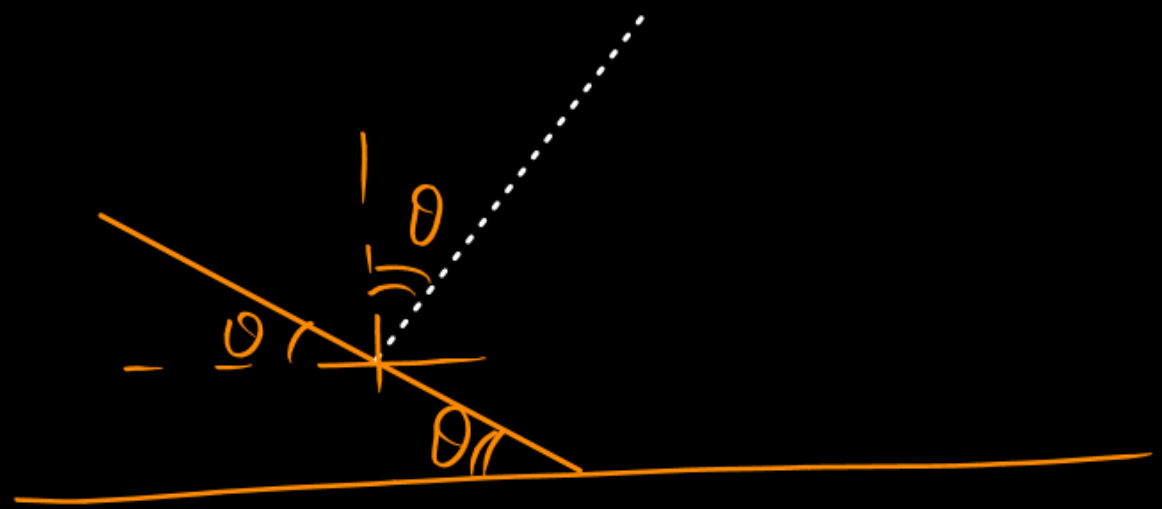
Two cars A and B are travelling towards each other on a single-lane road at 24 m/s and 21 m/s respectively. They notice each other when 180 m apart and apply brakes simultaneously. They just succeed in avoiding collision, both stopping simultaneously at the same position. Assuming constant retardation for each car, the distance travelled by car A while slowing down is

An inclined plane mirror lies at the bottom of a large tank containing a liquid as shown in figure. A narrow monochromatic beam falls on surface of water at angle of incidence θ . Beam comes out of the liquid at C. If maximum angle of incidence θ for which light comes out liquid is $\sin^{-1} \left(\frac{\sqrt{3}}{K} \right)$ then find

value of K.



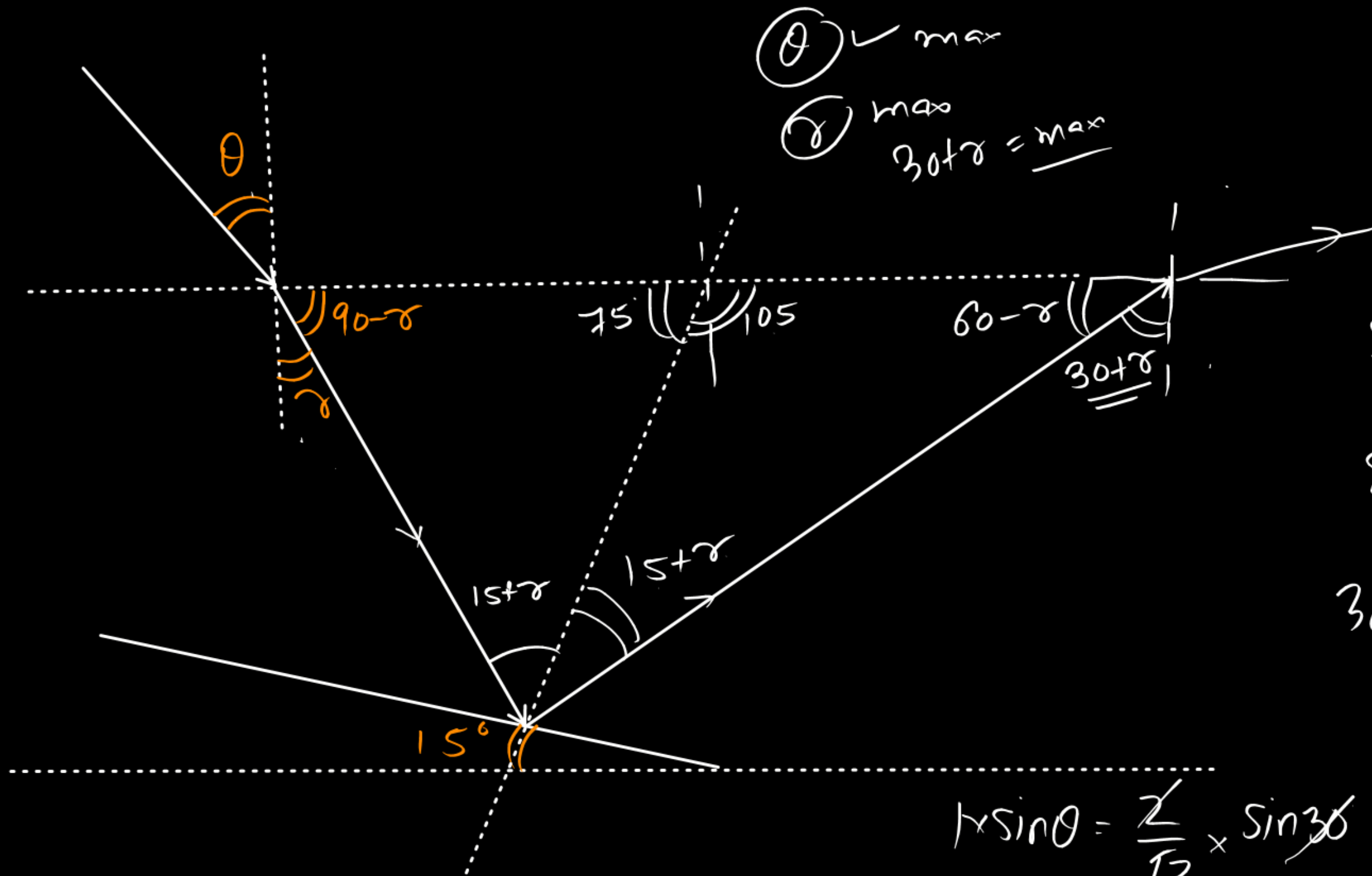
$$\theta = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$



$$\sin^{-1} \left(\frac{\sqrt{3}}{K} \right)$$

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{K}$$

$$K = 3$$



$$30 + \gamma \geq c$$

$$30 + \gamma = c$$

$$\sin(30 + \gamma) = \sin c = \frac{1}{2} \sqrt{3}$$

$$\sin(30 + \gamma) = \frac{\sqrt{3}}{2} = \sin 60$$

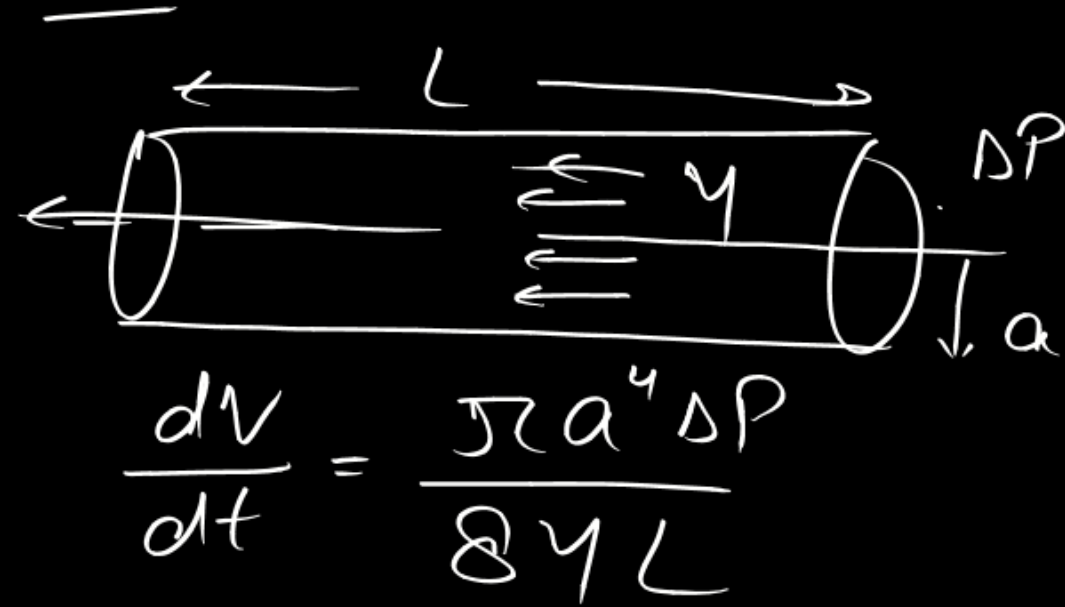
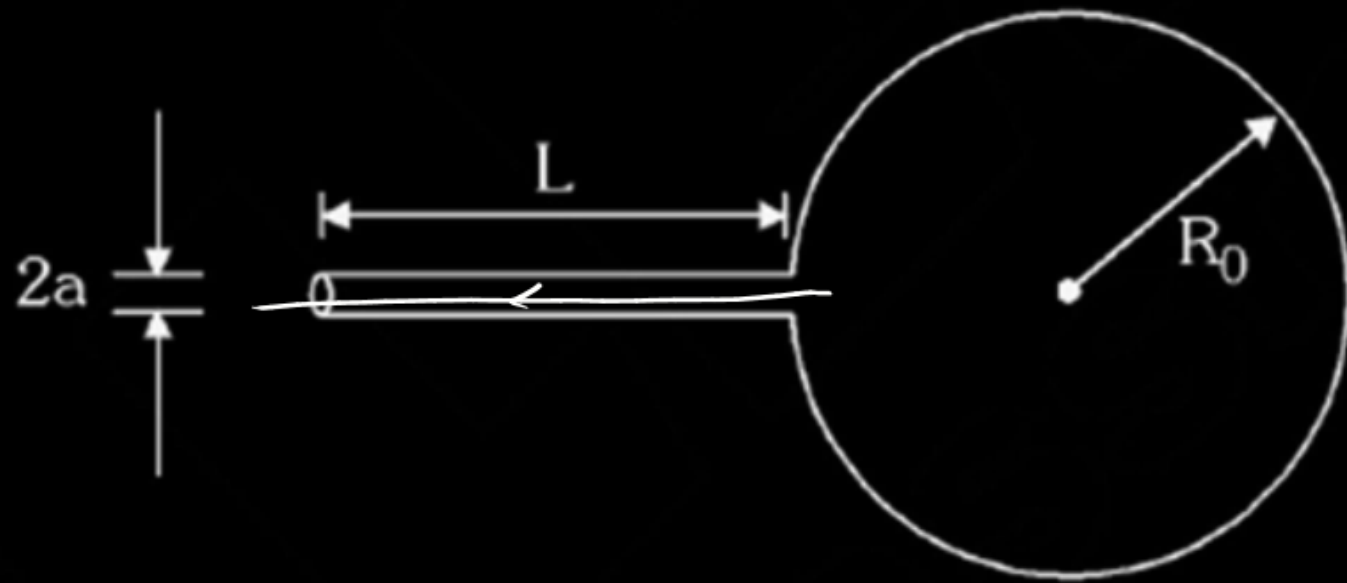
$$30 + \gamma = 60$$

$$\gamma = 30^\circ$$

$$1 \times \sin \theta = \frac{2}{\sqrt{3}} \times \sin 30 = \frac{1}{\sqrt{3}}$$

A soap bubble is blown at the end of a capillary tube of radius a and length L . When the other end is left open, the bubble begins to deflate. Surface tension of soap solution is T . It is known that volume flow rate through a tube of radius a and length L is given by Poiseuille's equation.

$$Q = \frac{\pi a^4 \Delta P}{8\eta L}$$

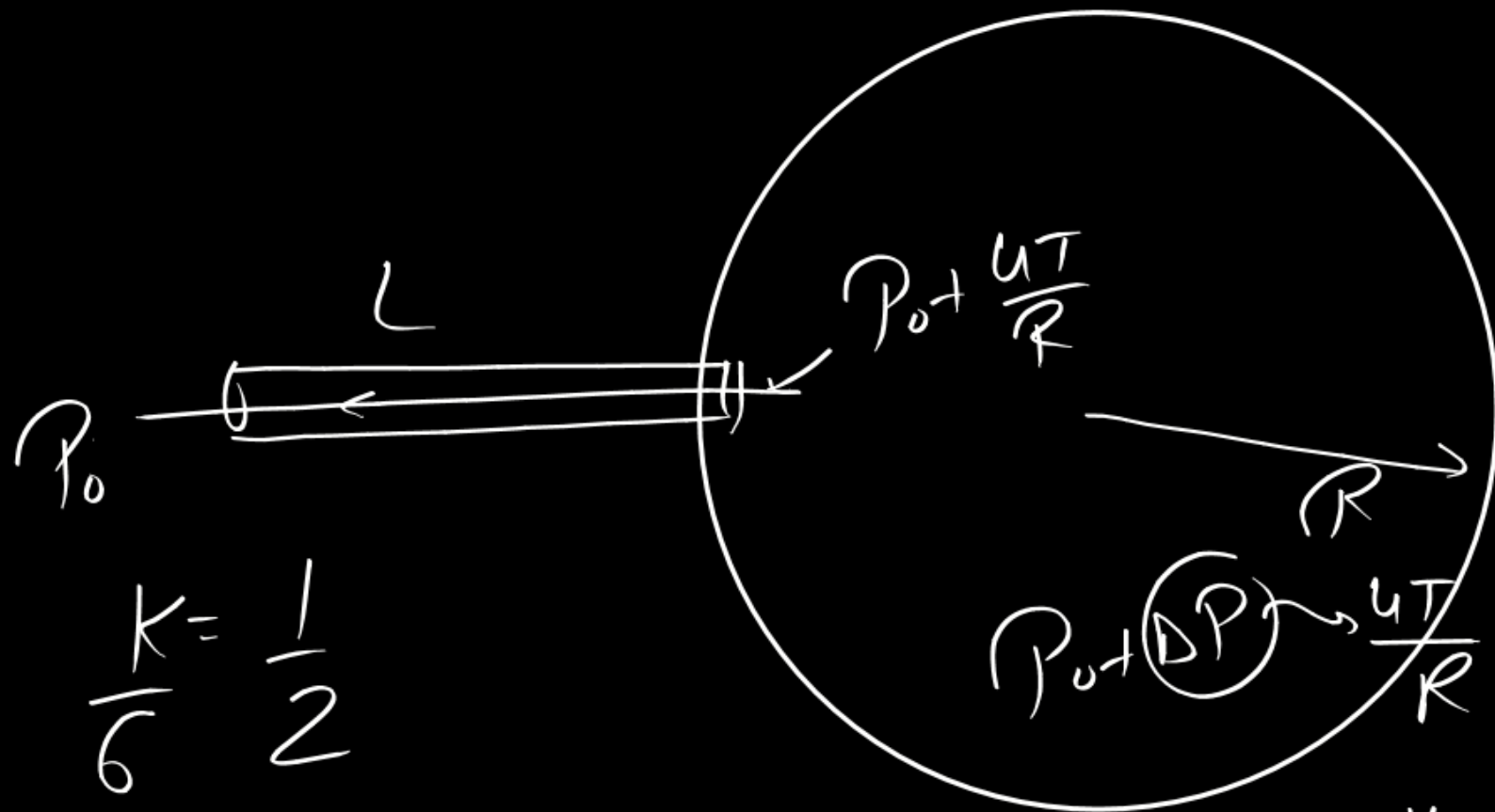


Where ΔP is pressure difference at the two ends of the tube and η is coefficient of viscosity. Assume that the bubble remains spherical. The initial radius of bubble was R_0 . The radius of bubble as a function of time t is found to be

$$R_0 \left[1 - \frac{K a^4 T \ell}{6 \eta L R_0^4} \right]^{1/4}. \text{ The value of } K \text{ is } \underline{\hspace{2cm}}.$$

$$R = R_0 \left\{ 1 - \frac{a^4 T t}{24 L R_0^4} \right\}^{1/4}$$

P_0



$$K = \frac{1}{2}$$

$$K = 3$$

$$\frac{dV}{dt} = \frac{\pi a^4 4T}{84 L R} = \frac{\pi a^4 T}{284 L R}$$

$$-\frac{a^4 T}{84 L} \int_0^t dt = \int_{R_0}^R R^3 dR$$

$$4\pi R^2 \frac{dR}{dt}$$

$$-\frac{a^4 T}{284 L} t = \frac{R^4 - R_0^4}{4}$$

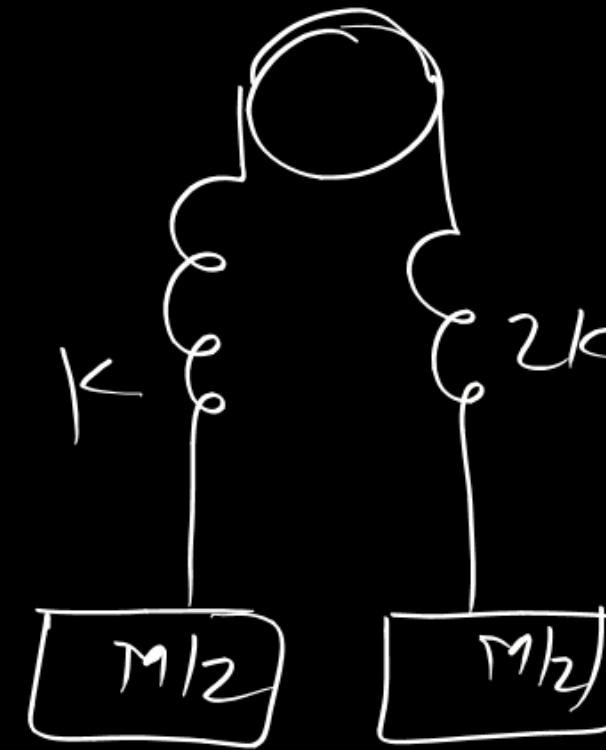
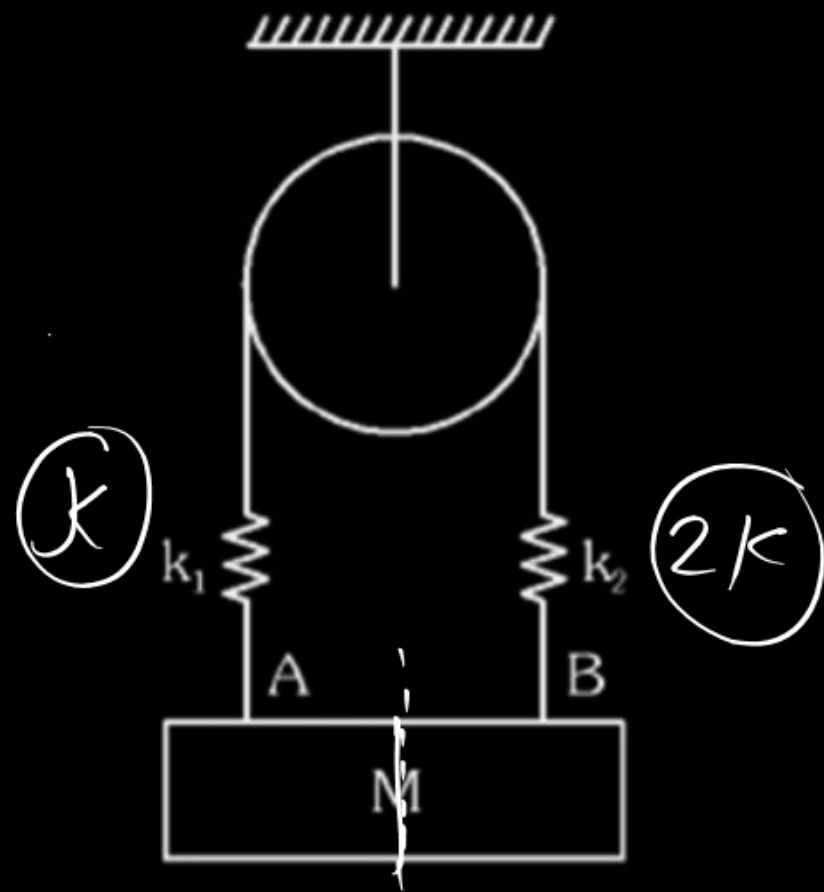
$$R^4 = \left(R_0^4 - \frac{a^4 T}{24 L} t \right) \Rightarrow R = \left(R_0^4 - \frac{a^4 T}{24 L} t \right)^{1/4}$$

$$V = \frac{4}{3} \pi R^3$$

$$\frac{dV}{dt} = \frac{4}{3} \pi R^2 \frac{dR}{dt}$$

$$= 4\pi R^2 \frac{dR}{dt}$$

In the system shown in the figure the string, springs and pulley are light. The force constant of the two springs are $k_1 = k_2$ and $k_2 = 2k$. Block of mass M is pulled vertically down from its equilibrium position and released. The top surface of the block (represented by line AB) always remains horizontal. The angular frequency of oscillation is found to be $\sqrt{\frac{Ck}{3m}}$. The value of C is _____.

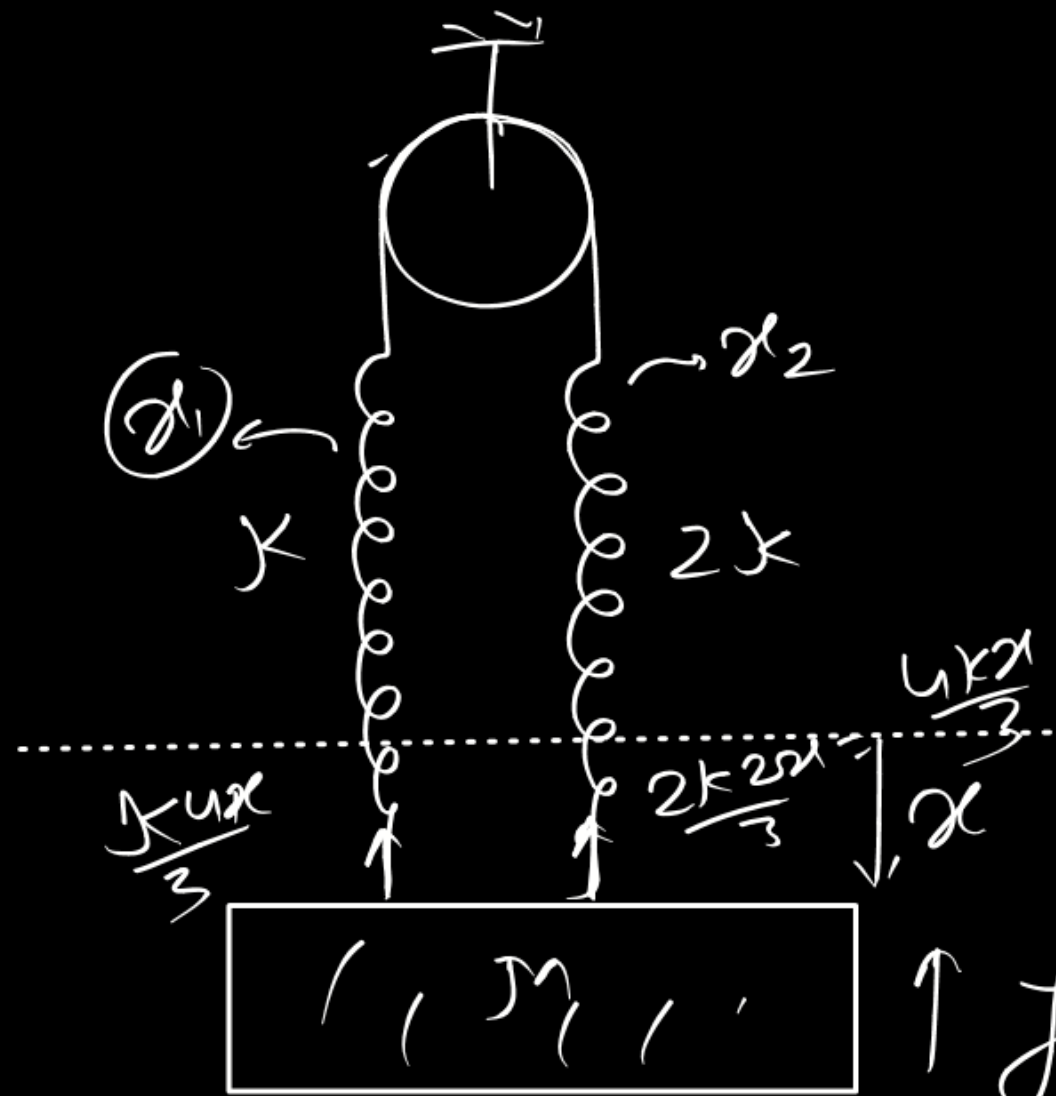


$$\frac{1}{k_e} = \frac{1}{k} + \frac{1}{2k} = \frac{1+2}{2k}$$

$$k_e = \frac{2k}{3}$$

$$T = 2\pi \sqrt{\frac{\mu}{k_e}} \quad \mu = \frac{m}{2} + \frac{m}{2} = \frac{m}{1}$$

$$\omega = \sqrt{\frac{k_e}{\mu}} = \sqrt{\frac{2k/3}{m}} = \sqrt{\frac{2k}{3m}}$$



$$kx_1 = 2kx_2$$

$$x_1 = 2x_2 \quad \text{--- (1)}$$

$$2x = x_1 + x_2$$

$$x_1 = \frac{4x}{3}$$

$$x_2 = \frac{2x}{3}$$

$$\uparrow F_{\text{net}} = \frac{8kx}{3} = ma$$

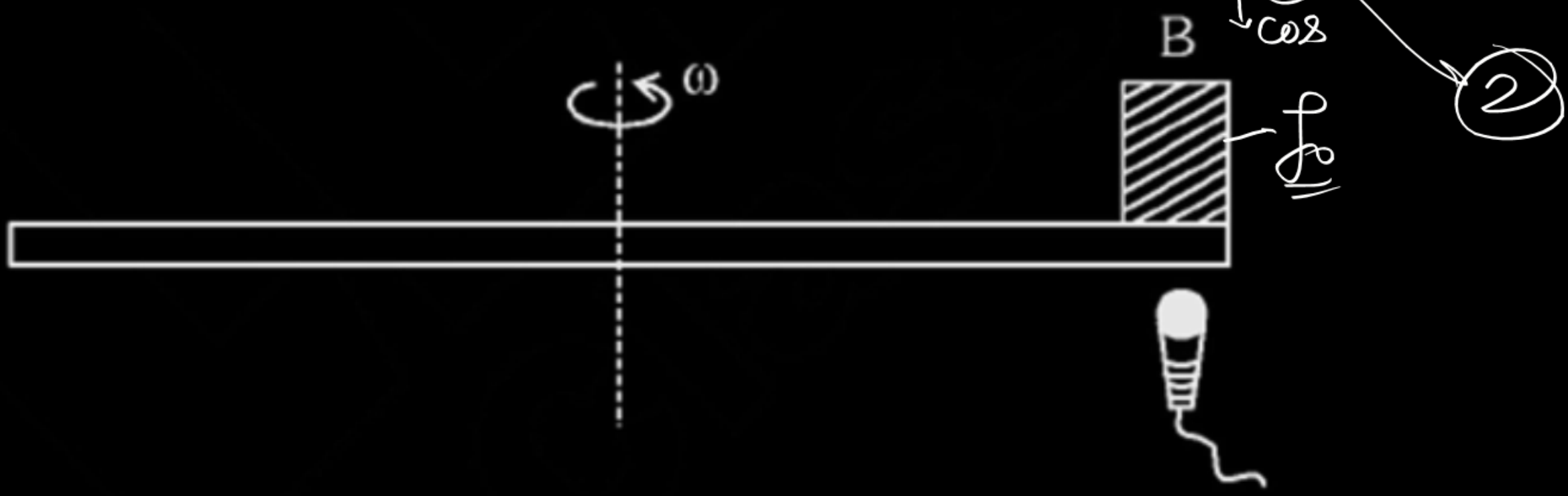
$$a = \left(\frac{8k}{3m} \right) x \rightarrow \omega^2$$

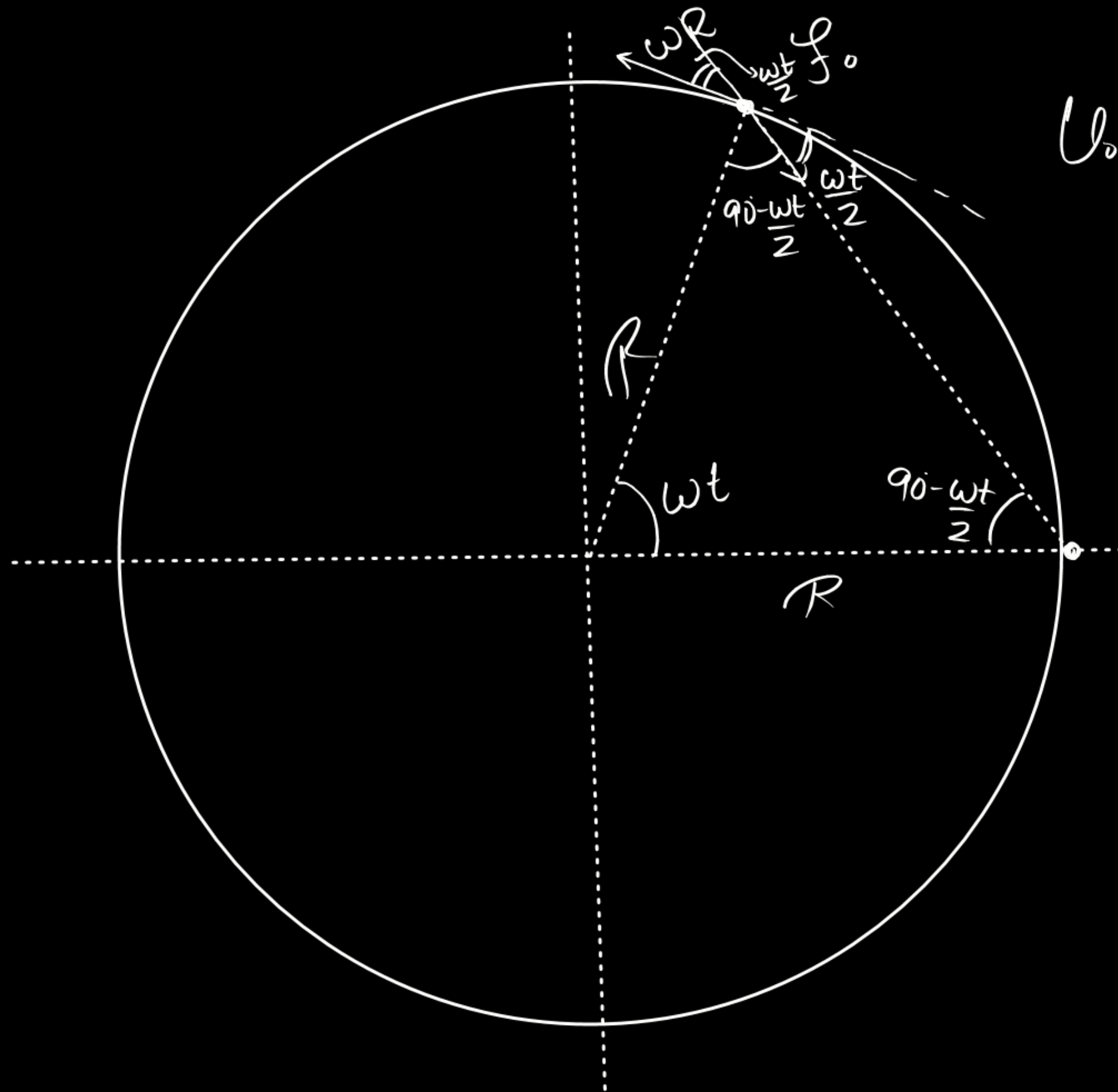
$$\omega = \sqrt{\frac{8k}{3m}} = \sqrt{\frac{Ck}{3m}}$$

$$C = 8$$

A wooden platform can be rotated about its vertical axis with constant angular speed ω with the help of a motor. A buzzer is fixed at the circumference of the platform and it rotates in a circle of radius R . The buzzer produces sound of frequency f_0 . A mic is placed just beneath the platform near its circumference. An electronic frequency analyzer connected to the mic records the frequency (f) received by the mic. Take (t) to be zero when the buzzer is just the mic. Speed of sound = V_0 . The

expression of f as a function of time is found to be $f_0 \left[\frac{V_0}{V_0 + R\omega \left(\frac{\omega t}{K} \right)} \right]$. The value of K is _____.





$$\omega R \cos \frac{\omega t}{2}$$

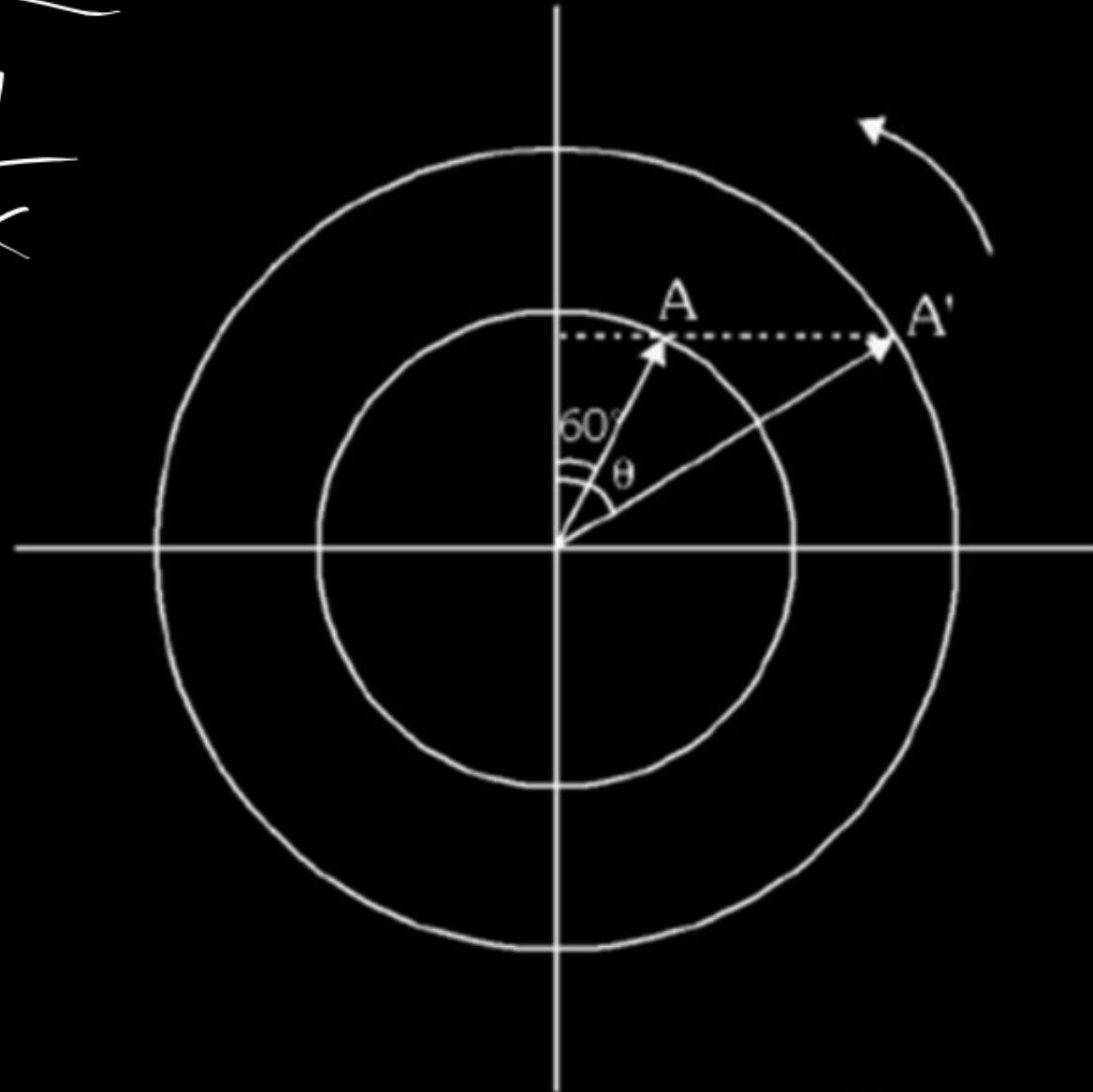
$$f_{app} = f_0 \left(\frac{U_0}{U_0 + \omega R \cos \frac{\omega t}{2}} \right)$$

A particle executing SHM with time period T and amplitude A along horizontal direction along x axis.

When it is at $+\frac{A}{2}$ from equilibrium position and moving towards right, it received an impulse which doubles its velocity without changing its direction. The time taken by particle to reach extreme right position is $\frac{T}{2\pi} \cos^{-1} \left[\frac{1}{K} \right]$. The value of K is _____.

$$K = \sqrt{13}$$

$$\frac{T}{2\pi} \cos^{-1} \frac{1}{K}$$



$$3A^2 = A'^2 - \frac{A^2}{4}$$

$$A' = \frac{\sqrt{13} A}{2}$$

$$A/2 = x$$

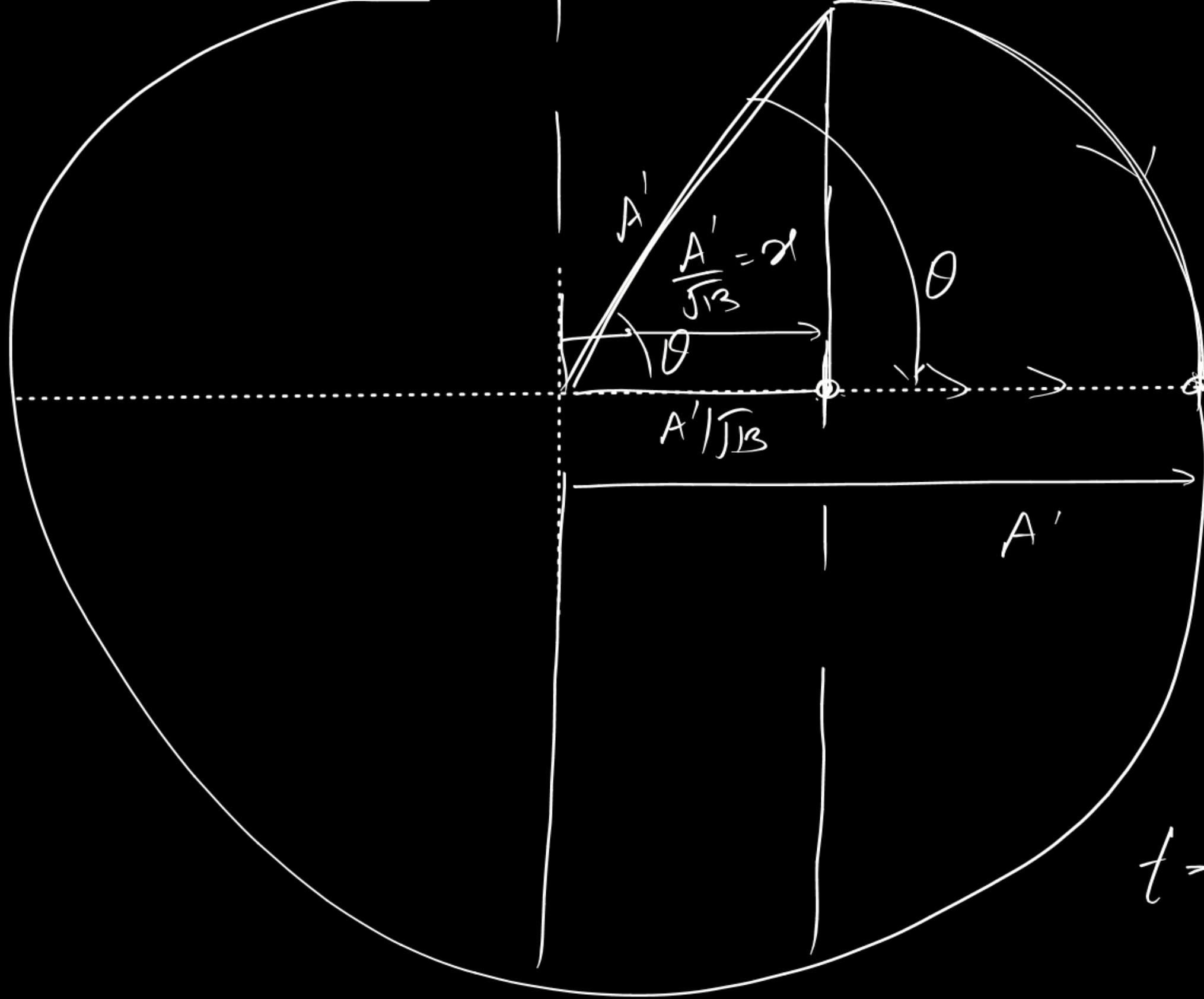
$$U = \omega \sqrt{A^2 - \frac{A^2}{4}}$$

$$= \frac{\sqrt{3} A \omega}{2}$$

$$A' = \frac{\sqrt{13} A}{2}$$

$$x = A/2$$

$$U = \sqrt{3} A \omega = \omega \sqrt{A'^2 - \frac{A^2}{4}}$$



$$\cos \theta = \frac{A'/\sqrt{3}}{A'}$$

$$= \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{3}} = \omega t$$

$$\cos^{-1} \frac{1}{\sqrt{3}} = \frac{2\pi}{T} t$$

$$t = \frac{T}{2\pi} \cos^{-1} \frac{1}{\sqrt{3}}$$